

Characteristics of Forces — Paper C

HARD • 50 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- A calculator is allowed, but every number on this paper is designed to work without one.
- Every answer must carry a **unit**, and every resultant force must carry a **direction**. An answer with no unit does not score the final mark.
- The mark for each question is shown in square brackets on the right.
- No formula is given on this paper. Values of g are given only where they are not Earth's.
- Several questions describe situations you may not have seen before — apply the ideas you already have.

SECTION 1 – THE SAME OBJECT, DIFFERENT GRAVITY

- 1.** The gravitational field strength on the Moon is 1.6 N/kg and on Mars it is 3.7 N/kg.

- (a)** A rover has a mass of 250 kg. Calculate its weight on Mars. [2]

- (b)** An instrument weighs 32 N on the Moon. Calculate its mass. [2]

- (c)** State the weight of that same instrument on Earth. [1]

2. A student writes: “My mass is 45 kg, so my weight is 45 kg.” Identify **two** separate [4]
errors in that sentence, and calculate the student’s correct weight on Earth.

SECTION 2 – FORCES IN MORE THAN ONE DIRECTION

3. A hot-air balloon is held by a rope. Its weight is 12 000 N downwards and the upthrust on it is 15 000 N upwards. A wind pushes it east with 900 N while the rope pulls it west with 900 N.

- (a) Calculate the resultant vertical force, and give its direction. [2]

- (b) Calculate the resultant horizontal force. [1]

- (c) Describe the motion of the balloon, in both directions. [2]

4. A box is dragged across a rough floor by a rope.

- (a) The rope pulls with 85 N and the box moves at a steady speed. State the size of [2]
the friction force and explain how you know.

- (b)** The pull is increased to 130 N while friction stays at 85 N. Calculate the resultant force and describe its effect on the box. [2]

SECTION 3 – JUDGEMENT

- 5.** Three forces act on a trolley along one straight line: 26 N to the right, 9 N to the left and 12 N to the left.

- (a)** Calculate the resultant force on the trolley. [2]

- (b)** A fourth force is now applied so that the trolley moves at a constant velocity. State the size and direction of that fourth force. [2]

- 6.** A skydiver of mass 80 kg jumps from a plane.

- (a)** Calculate her weight. [1]

- (b)** Just after she jumps, the air resistance acting on her is 150 N. Calculate the resultant force and state its direction. [2]

- (c)** Later she falls at a constant speed. State the air resistance at that moment. [1]

- 7.** A spacecraft far from any planet switches its engine off. It carries on at a steady speed in a straight line with nothing pushing it. Explain why. [4]

End of paper. Check every answer has a unit or a form the question actually asked for.